

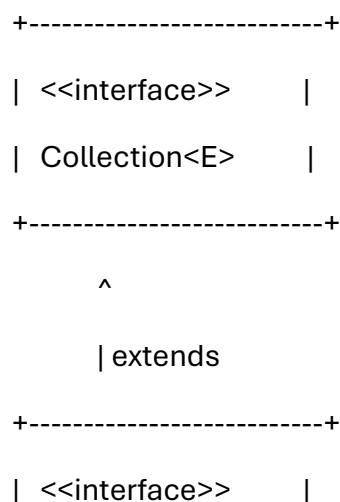
Aim

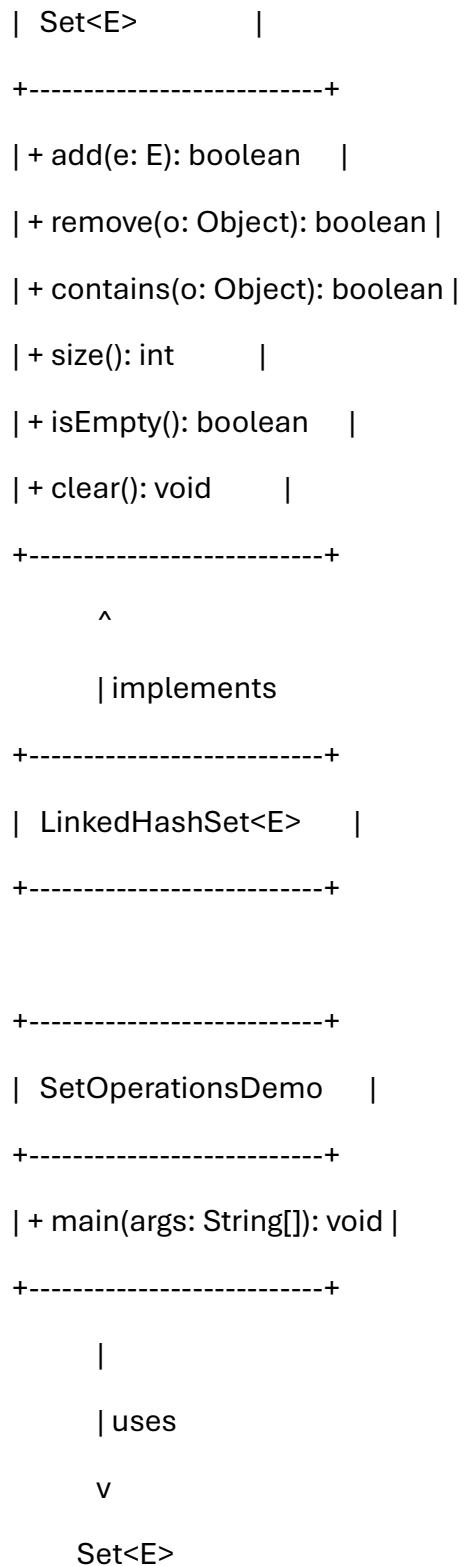
To write a Java program that demonstrates the basic operations of the Set interface, namely adding, removing and searching for elements, checking its size, testing whether it is empty, and clearing it, using the methods add(), remove(), contains(), size(), isEmpty() and clear(), and to show that a Set does not allow duplicate elements.

Algorithm:

1. Start.
2. Create a Set object of type String using LinkedHashSet (a HashSet works too, but its print order is not guaranteed).
3. Call isEmpty() to check that the new set is empty.
4. Call add() with "Apple", "Banana" and "Cherry". Each returns true.
5. Call add("Apple") again. It returns false because duplicates are not allowed.
6. Display the set and call size() to get the number of elements.
7. Call contains() with "Banana" and "Mango" to check for elements.
8. Call remove() with "Banana" and "Mango". It returns true if the element was present and false otherwise.
9. Display the set and its size again.
10. Call clear() to remove all elements, then call isEmpty() to confirm.
11. Stop.

UML Diagram:





Code:

```

package Observations;

import java.util.HashSet;

import java.util.Set;

```

```
public class SetInterface {  
    public static void main(String[] args) {  
        Set<String> set = new LinkedHashSet<>();  
  
        System.out.println("Is set empty? " + set.isEmpty());  
  
        System.out.println("add(\"Apple\"): " + set.add("Apple"));  
        System.out.println("add(\"Banana\"): " + set.add("Banana"));  
        System.out.println("add(\"Cherry\"): " + set.add("Cherry"));  
        System.out.println("add(\"Apple\") again: " + set.add("Apple"));  
  
        System.out.println("Set after adding: " + set);  
        System.out.println("Size: " + set.size());  
  
        System.out.println("contains(\"Banana\"): " + set.contains("Banana"));  
        System.out.println("contains(\"Mango\"): " + set.contains("Mango"));  
  
        System.out.println("remove(\"Banana\"): " + set.remove("Banana"));  
        System.out.println("remove(\"Mango\"): " + set.remove("Mango"));  
  
        System.out.println("Set after removal: " + set);  
        System.out.println("Size: " + set.size());  
  
        set.clear();  
  
        System.out.println("After clear(): " + set);  
        System.out.println("Is set empty? " + set.isEmpty());  
    }  
}
```

```
}
```

Output:

```
bservations.SetInterface'  
Is set empty? true  
add("Apple"): true  
add("Banana"): true  
add("Cherry"): true  
add("Apple") again: false  
Set after adding: [Apple, Banana, Cherry]  
Size: 3  
contains("Banana"): true  
contains("Mango"): false  
remove("Banana"): true  
remove("Mango"): false  
Set after removal: [Apple, Cherry]  
Size: 2  
After clear(): []  
Is set empty? true  
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>
```

Result:

The program was executed successfully and the Set methods behaved as expected. Duplicate elements were rejected, and remove() returned false for an element that was not present.